# Chapter 20
# Wind Power Devices

This chapter presents wind speed and wind turbine models. Wind power is particularly interesting from the modelling viewpoint since it combines stochastic models (i.e., wind speed), mechanics (i.e., wind turbine), electrical machines, power electronics (i.e., VSC devices) and controls. For this reason, wind power models conclude this part dedicated to power system modelling.

The chapter is divided into two sections. Section 20.1 describes three wind speed models, namely the Weibull's distribution, a wind model composed of average speed, ramp, gust and turbulence and the normalized Mexican hat wavelet model. Section 20.2 describes three models of wind turbines, namely the constant speed wind turbine with squirrel-cage induction generator, the variable speed wind turbine with doubly-fed (wound rotor) asynchronous generator and the variable speed wind turbine with direct-drive synchronous generator.

## 20.1 Wind Speed Models

The best way to model the wind speed is by an historical series of measurement data. However, measures are not adequate for comparison and benchmarking. Thus, fictitious mathematical wind speed models are of interest.

The wind speed models described in this section are the Weibull's distribution, a composite model that includes average speed, ramp, gust and turbulence, and the Mexican hat wavelet model.

Table 20.1 defines all parameters used in wind speed models that are described in the following subsections. Air density $\rho$ at 15°C and standard atmospheric pressure is 1.225 kg/m$^3$, and depends on the altitude (e.g., at 2000 m $\rho$ is 20% lower than at the sea level).

Wind speed time sequences are calculated after solving the power flow and initializing wind turbine variables. To simulate the smoothing of high-frequency wind speed variations over the rotor surface, the actual wind speed

F. Milano: Power System Modelling and Scripting, Power Systems, pp. 435–456.
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values is filtered through low-pass filter before being used for computing the mechanical power of the wind turbine (see Figure 20.1):

$$\dot{v}_m = (\check{v}_w(t) - v_w)/T_w \tag{20.1}$$

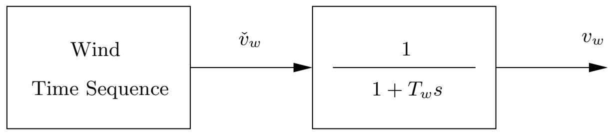


**Fig. 20.1** Low-pass filter to smooth wind speed variations

**Table 20.1** Wind speed parameters

| Variable | Description | Unit |
|---|---|---|
| $c_w$ | Scale factor for Weibull's distribution | - |
| $h_w$ | Height of the wind speed signal | m |
| $k_w$ | Shape factor for Weibull's distribution | - |
| $n_{\text{har}}$ | Number of harmonics | int |
| $t_0$ | Centering time of the Mexican hat wavelet | s |
| $t_e^g$ | Ending gust time | s |
| $t_e^r$ | Ending ramp time | s |
| $t_s^g$ | Starting gust time | s |
| $t_s^r$ | Starting ramp time | s |
| $T_w$ | Low-pass filter time constant | s |
| $v_w^g$ | Gust speed magnitude | m/s |
| $v_w^r$ | Ramp speed magnitude | m/s |
| $v_{wn}$ | Nominal wind speed | m/s |
| $z_0$ | Roughness length | m |
| $\Delta f$ | Frequency step | Hz |
| $\Delta t$ | Sample time for wind measurements | s |
| $\rho$ | Air density | kg/m$^3$ |
| $\sigma$ | Shape factor of the Mexican hat wavelet | - |

### *20.1.1 Weibull's Distribution*

A common way to describe the wind speed is by means of the Weibull's distribution, which is as follows:

$$f(v_w, c_w, k_w) = \frac{k_w}{c_w^k} v_w^{k-1} e^{-(\frac{v_w}{c_w})^{k_w}} \tag{20.2}$$

where $v_w$ is the wind speed and $c_w$ and $k_w$ are constants as defined in the wind model data matrix. Time variations $\xi_w(t)$ of the wind speed are then obtained by means of a Weibull's distribution, as follows:

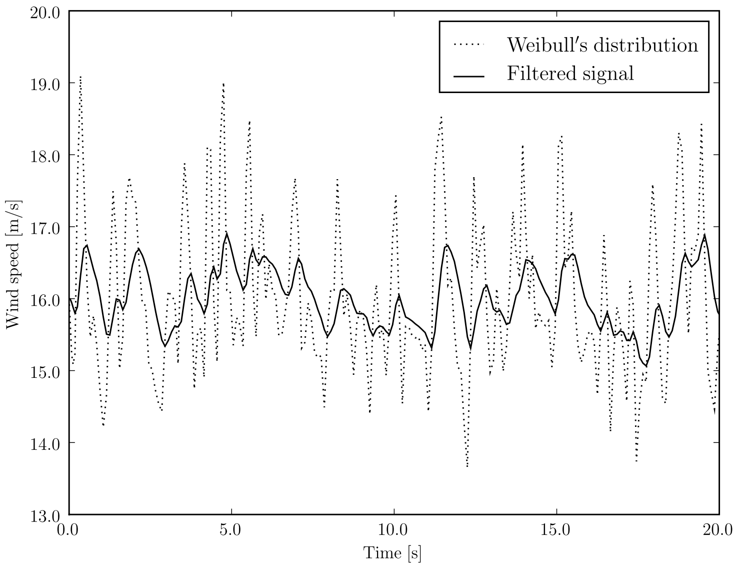


**Fig. 20.2** Weibull's distribution model of the wind speed

$$\xi_w(t) = \left(-\frac{\ln\iota(t)}{c_w}\right)^{\frac{1}{k_w}} \tag{20.3}$$

where $\iota(t)$ is a generator of random numbers ($\iota \in [0,1]$). Usually the shape factor $k_w = 2$, which leads to the Rayleigh's distribution, while $k_w > 3$ approximates the normal distribution and $k_w = 1$ gives the exponential distribution. The scale factor $c$ should be chosen in the range $c_w \in (1, 10)$. Finally, the wind speed is computed setting the initial average speed $v_w^a$ determined at the initialization step as mean speed:

$$\check{v}_w(t) = (1 + \xi_w(t) - \xi_w^a)v_w^a \tag{20.4}$$

where $\xi_w^a$ is the average value of $\xi_w(t)$.

### Example 20.1 Weibull's Distribution

An example of wind speed time sequence generated using a Weibull's distribution is depicted in Figure 20.2. Wind data are $v_{wn} = 16$ m/s, $\Delta t = 0.1$ s, $T_w = 0.5$ s, $c_w = 20$ and $k_w = 2$.

### 20.1.2 Composite Wind Speed Model

This subsection describes a composite wind model similar to what proposed in [343] and [9]. This model considers that the wind speed is composed of four parts:

1. Average and initial wind speed $v_w^a$.
2. Ramp component of the wind speed $v_w^r$.
3. Gust component of the wind speed $v_w^g$.
4. Wind speed turbulence $v_w^t$.

The resulting wind speed $\check{v}_w$ is:

$$\check{v}_w(t) = v_w^a + v_w^r(t) + v_w^g(t) + v_w^t(t) \tag{20.5}$$

where all components are time-dependent except for the average speed $v_w^a$.

#### Wind Ramp Component

The wind ramp component is defined by an amplitude $A_w^r$ and starting and ending times, $t_s^r$ and $t_e^r$ respectively:

$$\begin{aligned} t < t_s^r:&\ \ v_w^r(t) = 0 \\ t_s^r \le t \le t_e^r:&\ \ v_w^r(t) = A_w^r\left(\frac{t - t_s^r}{t_e^r - t_s^r}\right) \\ t > t_e^r:&\ \ v_w^r(t) = A_w^r \end{aligned} \tag{20.6}$$

#### Wind Gust Component

The wind gust component is defined by an amplitude $A_w^g$ and starting and ending times, $t_s^g$ and $t_e^g$ respectively:

$$\begin{aligned} t < t_s^g:&\ \ v_w^g(t) = 0 \\ t_s^g \le t \le t_e^g:&\ \ v_w^g(t) = \frac{A_w^g}{2}\left(1 - \cos\left(2\pi\frac{t - t_s^g}{t_e^g - t_s^g}\right)\right) \\ t > t_e^g:&\ \ v_w^g(t) = A_w^g \end{aligned} \tag{20.7}$$

#### Wind Turbulence Component

The wind turbulence component is described by a power spectral density, as follows:

$$S_w^t = \frac{\frac{1}{(\ln(h_w/z_0))^2}\ell v_w^a}{\left(1 + 1.5\frac{\ell f}{v_w^a}\right)^{\frac{5}{3}}} \tag{20.8}$$

where $f$ is the frequency, $h_w$ the wind turbine tower height, $z_0$ is the roughness length and $\ell$ is the turbulence length scale:

$$\begin{aligned} h_w < 30: \;\; \ell = 20h \\ h_w \geq 30: \;\; \ell = 600 \end{aligned} \tag{20.9}$$

Table 20.2 depicts roughness values $z_0$ for a variety of ground surfaces.

**Table 20.2** Roughness length $z_0$ for a variety of ground surfaces [231, 281]

| Ground surface | Roughness length $z_0$ [m] |
|---|---|
| Open sea, sand | $10^{-4} \div 10^{-3}$ |
| Snow surface | $10^{-3} \div 5 \cdot 10^{-3}$ |
| Mown grass, steppe | $10^{-3} \div 10^{-2}$ |
| Long grass, rocky ground | $0.04 \div 0.1$ |
| Forests, cities, hilly areas | $1 \div 5$ |

The spectral density is then converted in a time domain cosine series as illustrated in [283]:

$$v_w^t(t) = \sum_{i=1}^{n_{\mathrm{har}}} \sqrt{S_w^t(f_i)\Delta f} \cos(2\pi f_i t + \phi_i + \Delta\phi) \tag{20.10}$$

where $f_i$ and $\phi_i$ are the frequency and the initial phase of the $i^{\mathrm{th}}$ frequency component, being $\phi_i$ random phases ($\phi_i \in [0, 2\pi)$). The frequency step $\Delta f$ should be $\Delta f \in (0.1, 0.3)$ Hz. Finally $\Delta\phi$ is a small random phase angle introduced to avoid periodicity of the turbulence signal.

**Example 20.2 Composite Wind Model**

An example of wind speed time series generated using a composite model is depicted in Figure 20.3. Wind data are $v_{wn} = 16$ m/s, $\Delta t = 0.1$ s, $T_w = 0.5$ s, $t_s^r = 5$ s, $t_e^r = 15$ s, $v_w^r = 0.5$ m/s, $t_s^g = 5$ s, $t_e^g = 15$ s and $v_w^g = 0.2$ m/s, $h_w = 50$ m, $z_0 = 0.01$ m, $\Delta f = 0.1$ Hz and $n = 50$.

### *20.1.3 Mexican Hat Wavelet Model*

The Mexican hat wavelet is a deterministic wind gust that has been standardized by IEC [139]. The Mexican hat wavelet is the normalized second derivative of the Gauss' distribution function, i.e., up to scale normalization, the second Hermite's function, as follows:

$$\check{v}_w(t) = v_w^a + (v_w^g - v_w^a)\Big(1 - \frac{(t - t_0)^2}{\sigma^2}\Big) e^{-\frac{(t-t_0)^2}{2\sigma^2}} \tag{20.11}$$

where $t_0$ is the centering time of the gust, $\sigma$ is the gust shape factor, $v_w^g$ is the peak wind speed value and $v_w^a$ the average speed value.

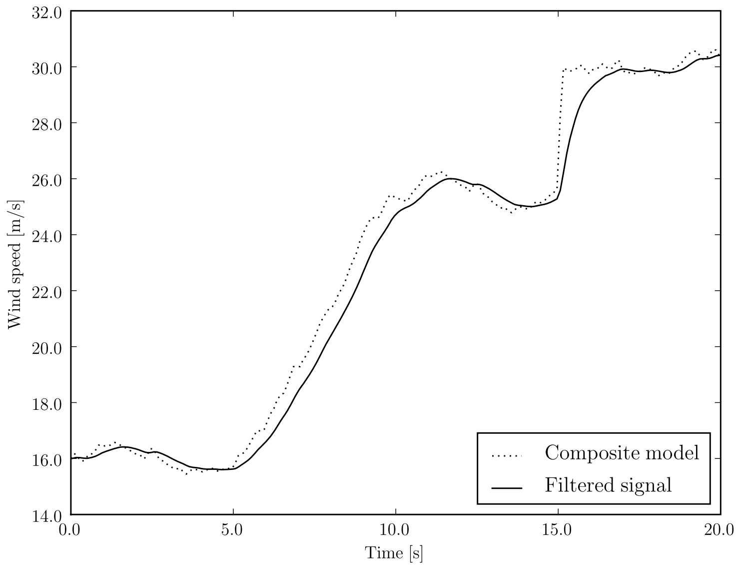


**Fig. 20.3** Composite model of the wind speed

**Example 20.3 Mexican Hat Wavelet Wind Model**

Figure 20.4 shows an example of wind gust modeled through a Mexican hat wavelet with $v_{wn} = 16$ m/s, $v_w^a = 12$ m/s, $\Delta t = 0.1$ s, $T_w = 0.5$, $t_0 = 10$ s, $\sigma = 1$, and $v_w^g = 25$ m/s.

## 20.2 Wind Turbines

This section describes the three most common wind turbine types: the non-controlled speed wind turbine with squirrel-cage induction generator, the controlled speed wind turbine with doubly-fed (wound rotor) asynchronous generator and the direct-drive synchronous generator [4]. Figure 20.5 depicts three wind turbines types, while Table 20.3 illustrates a few recent wind turbines data as documented in [91].

The squirrel-cage induction generator type is the oldest one. It has the advantages of being relatively cheap and electrically efficient. However, due to the lack of speed regulation, it is not aerodynamically efficient. This kind of wind generators is also noisy, requires a gearbox and the shaft suffers mechanical stresses (i.e., oscillations due the shadow effect and blade torsional modes). It has to be noted that the speed control is theoretically possible for this machines. However, this would require refurbishing existing turbines.

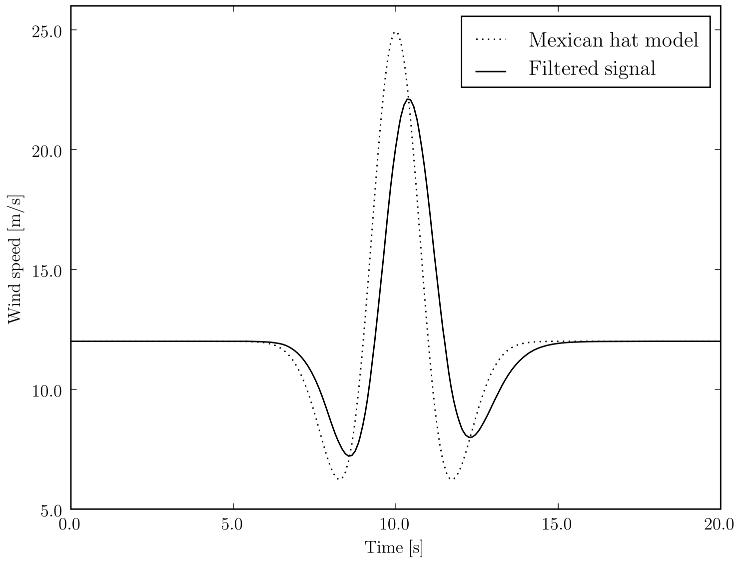


**Fig. 20.4** Mexican hat model of the wind speed

The doubly-fed asynchronous generator and direct-drive synchronous generator are aerodynamically more efficient than the squirrel-cage induction generator type thanks to the speed control. However, the electrical efficiency is lower. The need of two VSC converters with a back-to-back connection increases the cost but provides the ability of controlling the voltage and the power output. In the case of the synchronous generator, the converters are relatively more expensive than for the asynchronous generator because have to stand the entire generator power output. This fact has limited the diffusion the the direct-drive type. However, VSC converter cost is rapidly decreasing, hence the increasing interest in synchronous generators for wind power applications. Finally, no gearbox is required for the synchronous generator since the VSC converters fully decouple the machine from the grid.

## 20.2.1 Single Machine and Aggregate Models

The wind turbine and generators models that are described in following subsections are adequate for a single machine as well as for a wind park composed of several generators (i.e., aggregate wind turbine model). The rules for a correct definition of the wind turbine data are:

1. The nominal power $S_n$ is the total power in MVA of the single or aggregate wind turbine. If the wind turbine is an aggregate equivalent model of a

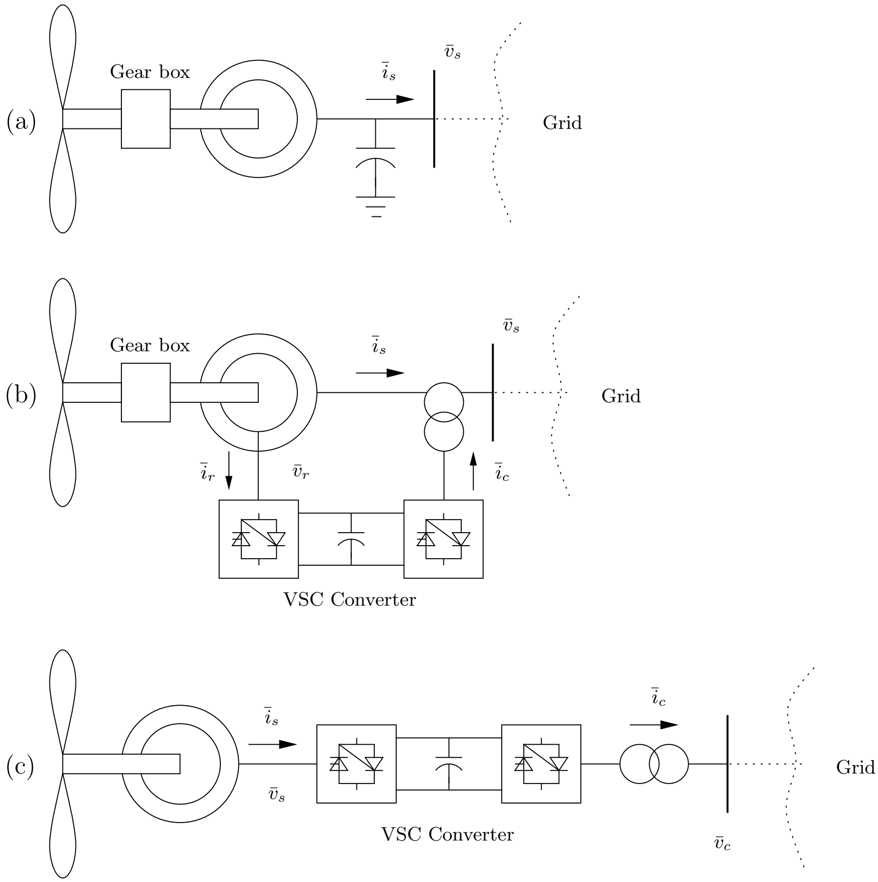


**Fig. 20.5** Wind turbine types. (a) Non-controlled speed wind turbine with squirrel-cage induction generator; (b) Controlled speed wind turbine with doubly-fed asynchronous generator; (c) Controlled speed wind turbine with direct-drive synchronous generator

**Table 20.3** Recent wind turbines [91]

| Type | Power [MW] | Diam. [m] | Height [m] | Control | Speed [rpm] |
|---|---|---|---|---|---|
| Bonus | 2 | 86 | 80 | GD/TS/PS | 17 |
| NEC NM 1500/72 | 1.5 | 72 | 98 | GD/TS/PS | 17.3 |
| Nordex N-80 | 2.5 | 80 | 80 | GD/VS/PC | 19 |
| Vestas V-80 | 2 | 80 | 78 | GD/VS/PC | 19 |
| Enercon e-66 | 1.5 | 66 | 85 | GD/VS/PC | 22 |

GD gearbox drive DD direct drive VS variable speed
TS two speed PC pitch control PS shift pitch by stall

wind park, then the number of generators $n_{\mathrm{gen}} > 1$. The capacity of each generator is about $[0.5, 5]$ MVA (being 1 and 2 MVA the most common capacities), thus $n_{\mathrm{gen}} \in [0.2 S_n, 2 S_n]$. An inconsistent value of $n_{\mathrm{gen}}$ can lead to an inadequate initialization of the wind speed.

2. Machine impedances and inertiæ are equivalent weighted pu values of the machines that compose the aggregate wind park.

### 20.2.2 Wind Turbine Initialization

Wind turbines are initialized after power flow analysis and a static generator is needed to impose the desired voltage and active power at the wind turbine bus. Once the power flow solution is available, $v_0$, $\theta_0$, $p_0$ and $q_0$ at the generation bus are used for initializing the state and input variables, the latter being the wind speed $v_{w0}$, which is used as the average wind speed $v_w^a$ for the wind speed models. Due to the nonlinearity of generator, converter and turbine models, the initialization requires the implementation of a Newton's method. For the interested reader, further insights on wind turbine initialization are given in [285] and [133].

The following subsections are organized taking into account implementation issues. The models that are common to all wind turbine are described first. These are the turbine model and the shaft model. Then, each specific wind generator type is described. With this aim, electrical machine and VSC converter models are described together.

### 20.2.3 Turbine Model

The mechanical model of the wind turbine is independent from the generator configuration. Thus, the mechanical equations of the turbine can be implemented as a separate class and then imported in the wind generator model.

Following items describe two models, namely fixed-blade and variable pitch angle position blade turbines. Table 20.4 defines all parameters used below.

**Table 20.4** Turbine mechanical parameters

| Variable | Description | Unit |
|---|---|---|
| $K_p$ | Pitch control gain | rad/pu |
| $n_{\mathrm{blade}}$ | Number of blades | int |
| $n_{\mathrm{gen}}$ | Number of machines that compose the wind park | int |
| $n_{\mathrm{pole}}$ | Number of poles | int |
| $S_n$ | Power rating | MVA |
| $R$ | Rotor radius | m |
| $T_p$ | Pitch control time constant | s |
| $\eta_{GB}$ | Gear box ratio | - |
| $\rho$ | Air density | kg/m$^3$ |

### Fixed-Blade Turbine Model

This model assumes fixed turbine blades and is adequate for wind generators without speed regulation. The mechanical power $p_w$ extracted from the wind is a function of the wind speed $v_w$ and the turbine rotor speed $\omega_t$ and can be approximated as follows:

$$p_w = \frac{n_{\mathrm{gen}}\rho}{2S_n} c_p(\lambda) A_r v_w^3 \tag{20.12}$$

where $c_p$ is the performance coefficient or power coefficient, $\lambda$ the tip speed ratio and $A_r = \pi R^2$ the area swept by the rotor. The speed tip ratio $\lambda$ is the ratio between the blade tip speed $v_{bt}$ and the wind upstream the rotor $v_w$:

$$\lambda = \frac{v_{bt}}{v_w} = \eta_{GB} \frac{2R\omega_t}{n_{\mathrm{pole}} v_w} \tag{20.13}$$

Finally, the $c_p(\lambda)$ curve is approximated as follows:

$$c_p = 0.44 \left( \frac{125}{\lambda_i} - 6.94 \right) e^{-\frac{16.5}{\lambda_i}} \tag{20.14}$$

with

$$\lambda_i = \frac{1}{\frac{1}{\lambda} + 0.002} \tag{20.15}$$

### Turbine Model with Pitch Angle Control

In this model, turbine blades can rotate in order to reduce the rotor speed in case of super-synchronous conditions. The angular position $\theta_p$ of the blades is called *pitch angle*. This turbine model is adequate for wind generators with speed control.

The mechanical power $p_w$ extracted from the wind is a function of the wind speed $v_w$, the rotor speed $\omega_m$ and the pitch angle $\theta_p$. The mechanical power $p_w$ can be approximated as:

$$p_w = \frac{n_{\mathrm{gen}}\rho}{2S_n} c_p(\lambda, \theta_p) A_r v_w^3 \tag{20.16}$$

where parameters and variables are the same as in (20.12) and the speed tip ratio $\lambda$ is defined in (20.13). The $c_p(\lambda, \theta_p)$ curve is approximated as follows [126]:

$$c_p = 0.22 \left( \frac{116}{\lambda_i} - 0.4\theta_p - 5 \right) e^{-\frac{12.5}{\lambda_i}} \tag{20.17}$$

with

$$\frac{1}{\lambda_i} = \frac{1}{\lambda + 0.08\theta_p} - \frac{0.035}{\theta_p^3 + 1} \tag{20.18}$$

Alternative equations are given in [284], as follows:

$$c_p = 0.73\left(\frac{151}{\lambda_i} - 0.58\theta_p - 0.002\theta^{2.14} - 13.2\right)e^{-\frac{18.4}{\lambda_i}} \tag{20.19}$$

with

$$\frac{1}{\lambda_i} = \frac{1}{\lambda - 0.02\theta_p} - \frac{0.003}{\theta_p^3 + 1} \tag{20.20}$$

As discussed above, the pitch angle $\theta_p$ is controlled to avoid super-synchronous speeds. The control diagram is shown in Figure 20.6 and described by the differential equation:

$$\dot{\theta}_p = (K_p\phi(\omega_m - \omega^{\text{ref}}) - \theta_p)/T_p \tag{20.21}$$

where $\phi$ is a function which allows varying the pitch angle set point only when the difference $(\omega_m - \omega^{\text{ref}})$ exceeds a predefined value $\pm\Delta\omega$. Since the pitch control works only for super-synchronous speeds, an anti-windup limiter locks the pitch angle to $\theta_p = 0$ for sub-synchronous speeds.

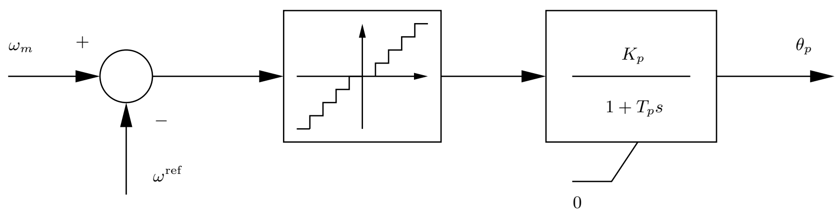


**Fig. 20.6** Pitch angle control diagram

The speed control is aimed to maximize the power production of the wind turbine. Figure 20.7 shows the dependence of the mechanical power $p_w$ produced by the wind turbine on the wind speed $v_w$ and the turbine rotor speed $\omega_t$. The solid line is the maximum mechanical power locus for each wind and rotor speeds. This curve is used for defining, for each value of the rotor speed, the optimal mechanical power $p_w^*$ that the turbine has to produce. Figure 20.8 shows a possible implementation of the $(p_w^*, \omega_t)$ characteristic. For super-synchronous speeds, the reference power is fixed to 1 pu to avoid overloading the generator. For $\omega_t < 0.5$ pu, the reference mechanical power is set to zero.

The detailed $(p_w^*, \omega_t)$ characteristic is:

$$p_w^*(\omega_t) = \begin{cases} 0 \quad \text{if} \quad \omega_t < 0.5 \\ p_w(\omega_t^*) \ \text{ that satisfies } \ \dfrac{dp_w(\omega_t^*, v_w, \theta_p)}{d\omega_t} = 0 \quad \text{if} \quad 0.5 \leq \omega_t \leq 1 \\ 1 \quad \text{if} \quad \omega_m > 1 \end{cases} \tag{20.22}$$

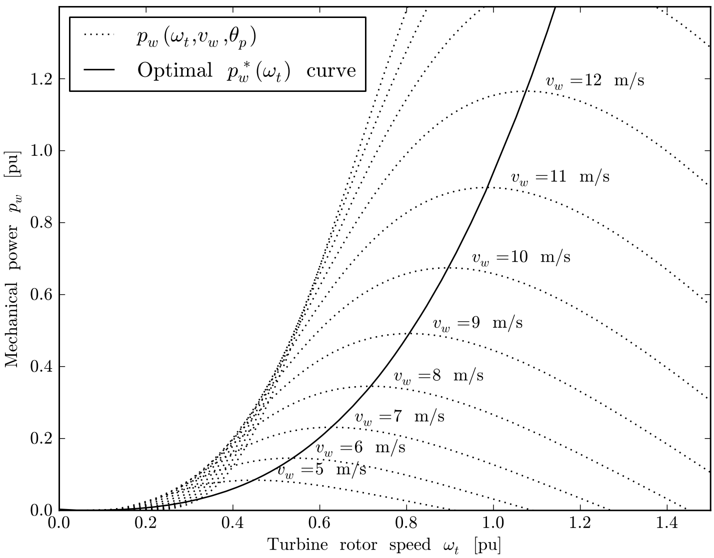


**Fig. 20.7** Speed-power characteristic of the wind turbine. The pitch angle is assumed $\theta_p = 0$ to plot the $p_w(\omega_t, v_w, \theta_p)$ curve

The simplified $(p_w^*, \omega_t)$ characteristic is:

$$p_w^*(\omega_t) = \begin{cases} 0 \quad \text{if} \quad \omega_t < 0.5 \\ 2\omega_t - 1 \quad \text{if} \quad 0.5 \leq \omega_t \leq 1 \\ 1 \quad \text{if} \quad \omega_t > 1 \end{cases} \tag{20.23}$$

## 20.2.4 Dynamic Shaft

The shaft of wind generators can be modelled as two masses connected by a spring. The masses represent the turbine shaft and the generator shaft, while the spring models the shaft stiffness. The resulting model is similar to the one described in Section 15.1.10 of Chapter 15.

Mechanical differential equations are:

$$\begin{aligned} \dot{\omega}_t &= (\tau_t - K_s \delta_{tm})/(2H_t) \\ \dot{\omega}_m &= (K_s \delta_{tm} - \tau_e)/(2H_m) \\ \dot{\delta}_{tm} &= \Omega_b(\omega_t - \omega_m) \end{aligned} \tag{20.24}$$

where $\Omega_b$ is the system rated frequency in rad/s, $\omega_t$ is the wind turbine angular speed, $\omega_m$ is the generator rotor speed, $\delta_{tm}$ is the relative angle

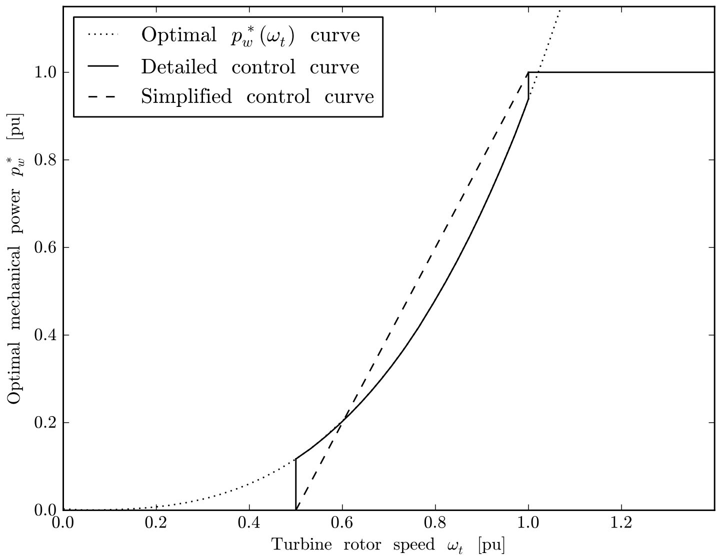


**Fig. 20.8** Optimal and implemented control speed-power characteristics

displacement of the two shafts, $\tau_e$ is the electrical torque and $\tau_t$ is the mechanical torque:

$$\tau_t = \frac{p_w}{\omega_t} \tag{20.25}$$

and all other parameters are defined in Table 20.5.

A periodic torque pulsation can be added to $\tau_t$ to simulate the tower shadow effect. The shadow-effect frequency depends on the rotor speed $\omega_t$, the gear box ratio $\eta_{GB}$, and the number of blades $n_{\text{blade}}$, as follows:

$$\tilde{\tau}_t = \tau_t\Big(1 + \alpha_s \sin\Big(\eta_{GB}\frac{\Omega_b \omega_t}{n_{\text{blade}}} t\Big)\Big) \tag{20.26}$$

where the torque pulsation amplitude can be fixed equal to $\alpha_s = 0.025$ [7]. The expression (20.26) substitutes the mechanical torque in the first equation of (20.24).

Shaft oscillations cannot be removed in the squirrel-cage induction generator type. For other wind turbine types that include VSC devices, the converter controls can effectively damp shadow effect modes and shaft oscillations [187]. If the control is efficient enough, the shaft can be considered rigid, i.e., $\omega_t = \omega_m$. Hence:

$$\dot{\omega}_t = (\tau_t - \tau_e)/(2H_t + 2H_m) \tag{20.27}$$

**Table 20.5** Wind turbine shaft parameters

| Variable | Description | Unit |
|---|---|---|
| $H_m$ | Machine rotor inertia constant | MWs/MVA |
| $H_t$ | Wind turbine inertia constant | MWs/MVA |
| $K_s$ | Shaft stiffness | pu |
| $\alpha_s$ | Shadow effect factor | - |

## *20.2.5 Non-Controlled Speed Wind Turbine*

The simplified electrical circuit used for the squirrel-cage induction generator is the same as the one for the single-cage induction motor, shown in Figure 15.14 of Chapter 15. The only difference with respect to the induction motor is that the currents are positive if injected into the network. The equations are formulated in terms of the real ($d$-) and imaginary ($q$-) axes, with respect to the network reference angle. Table 20.6 summarizes and defines all parameters of the squirrel-cage induction machine.

**Table 20.6** Squirrel-cage induction machine parameters

| Variable | Description | Unit |
|---|---|---|
| $r_r$ | Rotor resistance | pu |
| $r_s$ | Stator resistance | pu |
| $x_r$ | Rotor reactance | pu |
| $x_s$ | Stator reactance | pu |
| $x_\mu$ | Magnetizing reactance | pu |

### Network Interface

In a synchronously rotating reference frame, the link between the network and the stator machine voltages is:

$$\begin{aligned} v_d &= -v_h \sin\theta_h \\ v_q &= v_h \cos\theta_h \end{aligned} \tag{20.28}$$

and the active and reactive power productions are:

$$\begin{aligned} p_h &= v_d i_d + v_q i_q \\ q_h &= v_q i_d - v_d i_q + b_c(v_d^2 + v_q^2) \end{aligned} \tag{20.29}$$

where $b_c$ is the fixed capacitor conductance which is determined at the initialization step to impose the required bus voltage level.

### Machine Electro-Magnetic Equations

The differential equations in terms of the voltage behind the stator resistance $r_S$ are:

$$\begin{aligned} e'_d - v_d &= r_s i_d - x' i_q \\ e'_q - v_q &= r_s i_q + x' i_d \end{aligned} \tag{20.30}$$

whereas the link between voltages, currents and state variables is as follows:

$$\begin{aligned} \dot{e}'_d &= \Omega_b(1-\omega_m)e'_q - (e'_d - (x_0 - x')i_q)/T'_0 \\ \dot{e}'_q &= -\Omega_b(1-\omega_m)e'_d - (e'_q + (x_0 - x')i_d)/T'_0 \end{aligned} \tag{20.31}$$

where $\omega_m$ is the rotor angular speed, and $x_0$, $x'$ and $T'_0$ can be obtained from generator parameters:

$$\begin{aligned} x_0 &= x_s + x_\mu \\ x' &= x_s + \frac{x_r x_\mu}{x_r + x_\mu} \\ T'_0 &= \frac{x_r + x_\mu}{\Omega_b r_R} \end{aligned} \tag{20.32}$$

### Turbine and Machine Mechanical Equations

The mechanical DAE system is (20.12) and (20.24)-(20.26), where the electrical torque $\tau_e$ is defined as:

$$\tau_e = e'_d i_d + e'_q i_q \tag{20.33}$$

## 20.2.6 Doubly-Fed Asynchronous Generator

The model of the doubly-fed asynchronous generator is assumed steady-state, as the stator and rotor flux dynamics are fast with respect to grid dynamics. As a result of these assumptions, one has the DAE system described below. Table 20.7 summarizes and defines all parameters required by wind turbine with doubly-fed asynchronous generator.

### Network Interface

Stator voltages depends on the bus voltage $\bar{v}_h$:

$$\begin{aligned} v_{s,d} &= -v_h \sin\theta_h \\ v_{s,q} &= v_h \cos\theta_h \end{aligned} \tag{20.34}$$

**Table 20.7** Doubly-fed asynchronous generator parameters

| Variable | Description | Unit |
|---|---|---|
| $K_V$ | Voltage control gain | pu/pu/s |
| $p^{\max}$ | Maximum active power | pu |
| $p^{\min}$ | Minimum active power | pu |
| $q^{\max}$ | Maximum reactive power | pu |
| $q^{\min}$ | Minimum reactive power | pu |
| $r_r$ | Rotor resistance | pu |
| $r_s$ | Stator resistance | pu |
| $T_\epsilon$ | Power control time constant | s |
| $x_r$ | Rotor reactance | pu |
| $x_s$ | Stator reactance | pu |
| $x_\mu$ | Magnetizing reactance | pu |

The generator active and reactive power productions depend on the stator and converter currents $i_{s,d}+ji_{s,q}$ and $i_{c,d}+ji_{c,q}$, respectively, and stator and converter voltages $v_{s,d}+jv_{s,q}$ and $v_{c,d}+jv_{c,q}$, respectively, as follows:

$$\begin{aligned} p_h &= v_{s,d}i_{s,d} + v_{s,q}i_{s,q} + v_{c,d}i_{c,d} + v_{c,q}i_{c,q} \\ q_h &= v_{s,q}i_{s,d} - v_{s,d}i_{s,q} + v_{c,q}i_{c,d} - v_{c,d}i_{c,q} \end{aligned} \tag{20.35}$$

The expressions above can be rewritten as a function of stator and rotor currents $i_{s,d}+ji_{s,q}$ and $i_{r,d}+ji_{r,q}$, respectively, and stator and rotor voltages $v_{s,d}+jv_{s,q}$ and $v_{r,d}+jv_{r,q}$, respectively. In fact, the converter powers on the grid side are:

$$\begin{aligned} p_c &= v_{c,d}i_{c,d} + v_{c,q}i_{c,q} \\ q_c &= v_{c,q}i_{c,d} - v_{c,d}i_{c,q} \end{aligned} \tag{20.36}$$

whereas, on the rotor side:

$$\begin{aligned} p_r &= v_{r,d}i_{r,d} + v_{r,q}i_{r,q} \\ q_r &= v_{r,q}i_{r,d} - v_{r,d}i_{r,q} \end{aligned} \tag{20.37}$$

Assuming a loss-less converter model, the active power of the converter coincides with the rotor active power, thus $p_c = p_r$. The reactive power injected into the grid can be approximated neglecting stator resistance and assuming that the $d$-axis coincides with the maximum of the stator flux. Therefore, the powers injected in the grid are:

$$\begin{aligned} p_h &= v_{s,d}i_{s,d} + v_{s,q}i_{s,q} + v_{r,d}i_{r,d} + v_{r,q}i_{r,q} \\ q_h &= -\frac{x_\mu v_h i_{r,d}}{x_s + x_\mu} - \frac{v_h^2}{x_\mu} \end{aligned} \tag{20.38}$$

### Machine Electro-Magnetic Equations

The machine stator and rotor voltages are a function of stator and rotor currents and the rotor speed $\omega_m$:

$$
\begin{aligned}
v_{s,d} &= -r_s i_{s,d} + ((x_s + x_\mu) i_{s,q} + x_\mu i_{r,q}) \\
v_{s,q} &= -r_s i_{s,q} - ((x_s + x_\mu) i_{s,d} + x_\mu i_{r,d}) \\
v_{r,d} &= -r_r i_{r,d} + (1 - \omega_m)((x_s + x_\mu) i_{r,q} + x_\mu i_{s,q}) \\
v_{r,q} &= -r_r i_{r,q} - (1 - \omega_m)((x_s + x_\mu) i_{r,d} + x_\mu i_{s,d})
\end{aligned}
\tag{20.39}
$$

whereas the links between stator fluxes and generator currents are:

$$
\begin{aligned}
\psi_{s,d} &= -((x_s + x_\mu) i_{s,d} + x_\mu i_{r,d}) \\
\psi_{s,q} &= -((x_s + x_\mu) i_{s,q} + x_\mu i_{r,q})
\end{aligned}
\tag{20.40}
$$

### Turbine and Machine Mechanical Equations

The generator motion equation is modeled as a single shaft, i.e., (20.27), as it is assumed that the converter controls are able to filter shaft dynamics. For the same reason, no tower shadow effect is considered in this model. In (20.27), the electrical torque is:

$$
\tau_e = \psi_{s,d} i_{s,q} - \psi_{s,q} i_{s,d} \tag{20.41}
$$

Substituting the stator flux equations (20.40) in (20.41) leads to:

$$
\tau_e = x_\mu (i_{r,q} i_{s,d} - i_{r,d} i_{s,q}) \tag{20.42}
$$

In [286], the following approximation of the electrical torque $\tau_e$ is proposed:

$$
\tau_e \approx -\frac{x_\mu v_h i_{r,q}}{\Omega_b (x_s + x_\mu)} \tag{20.43}
$$

where $\Omega_b$ is the system rated frequency in rad/s.

Mechanical equations are completed by the mechanical torque $\tau_t$ equation (20.25) and the turbine model (20.16) and (20.17)-(20.18) or (20.19)-(20.20) and the pitch angle control (20.21).

### VSC Regulators

Since VSC dynamics are quite fast with respect to the electro-mechanical transients, the converter can be modeled as an ideal current source, where $i_{r,q}$ and $i_{r,d}$ are state variables and are used for controlling the rotor speed and the bus voltage, respectively. VSC control diagrams are depicted in Figures 20.9 and 20.10. The DAE system of the converter currents is:

$$\dot{i}_{r,q} = \left(-\frac{x_s + x_\mu}{x_\mu v} p_w^*(\omega_m)/\omega_m - i_{r,q}\right)\frac{1}{T_\epsilon} \tag{20.44}$$
$$\dot{i}_{r,d} = K_V(v_h - v^{\mathrm{ref}}) - v_h/x_\mu - i_{r,d}$$

where $v^{\mathrm{ref}}$ is the reference voltage computed at the initialization step and $p_w^*(\omega_m)$ is the power-speed characteristic (20.22) or (20.23).

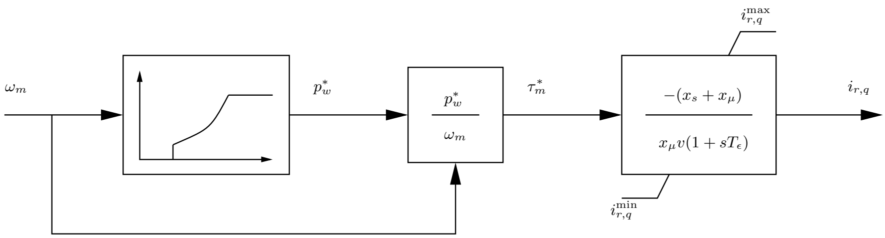


**Fig. 20.9** Rotor speed control diagram

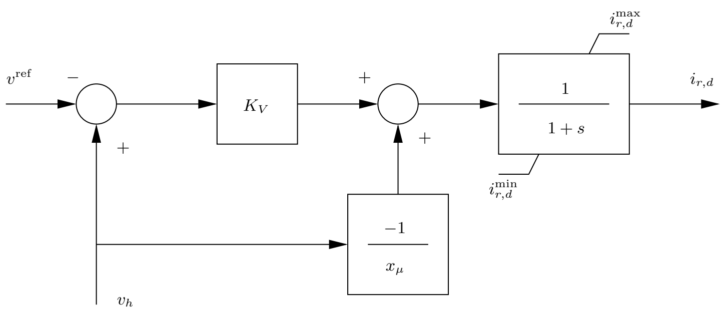


**Fig. 20.10** Voltage control diagram of the doubly-fed asynchronous generator

## Hard Limits

Both the speed and voltage controls undergo anti-windup limiters to avoid converter over-currents. Rotor current limits are computed based on active and reactive limits, and assuming bus voltage $v_h \approx 1$ one has:

$$\begin{aligned} i_{r,q}^{\max} &\approx -\frac{x_s + x_\mu}{x_\mu} p^{\min} \\ i_{r,q}^{\min} &\approx -\frac{x_s + x_\mu}{x_\mu} p^{\max} \\ i_{r,d}^{\max} &\approx -\frac{x_s + x_\mu}{x_\mu} q^{\min} - \frac{x_s + x_\mu}{x_\mu^2} \\ i_{r,d}^{\min} &\approx -\frac{x_s + x_\mu}{x_\mu} q^{\max} - \frac{x_s + x_\mu}{x_\mu^2} \end{aligned} \tag{20.45}$$

### 20.2.7 Direct-Drive Synchronous Generator

The model of the direct-drive synchronous generator is assumed steady-state, as the stator and rotor flux dynamics are fast with respect to grid dynamics. Furthermore, the converter decouples the generator from the grid. As a result of these assumptions, one has the DAE system described below. Table 20.8 summarizes and defines all parameters required by wind turbine with direct-drive synchronous generator.

**Table 20.8** Direct-drive synchronous generator parameters

| Variable | Description | Unit |
|---|---|---|
| $K_{dc}$ | Gain of the bus voltage control | pu/pu |
| $K_{ds}$ | Gain of the generator reactive power control | pu/pu |
| $K_{qc}$ | Gain of the active power control | pu/pu |
| $i^{\max}$ | Maximum current | pu |
| $r_s$ | Stator resistance | pu |
| $T_{dc}$ | Time constant of the bus voltage control | s |
| $T_{ds}$ | Time constant of the generator reactive power control | s |
| $T_{qc}$ | Time constant of the active power control | s |
| $T_{qs}$ | Time constant of the speed control | s |
| $x_d$ | $d$-axis reactance | pu |
| $x_q$ | $q$-axis reactance | pu |
| $\psi_p$ | Permanent field flux | pu |

#### Network Interface

The active and reactive powers injected into the grid depend on the grid side current $i_{c,d} + ji_{c,q}$ and voltage $v_{c,d} + jv_{c,q}$ of the converter:

$$\begin{aligned} p_h = p_c &= v_{c,d} i_{c,d} + v_{c,q} i_{c,q} \\ q_h = q_c &= v_{c,q} i_{c,d} - v_{c,d} i_{c,q} \end{aligned} \tag{20.46}$$

where the converter voltages are functions of the grid voltage magnitude and phase, as follows:

$$\begin{aligned} v_{c,d} &= -v_h \sin\theta_h \\ v_{c,q} &= v_h \cos\theta_h \end{aligned} \tag{20.47}$$

#### Machine Electro-Magnetic Equations

Assuming a permanent magnet synchronous generator, machine equations are

$$\begin{aligned} v_{s,d} &= -r_s i_{s,d} + \omega_m x_q i_{s,q} \\ v_{s,q} &= -r_s i_{s,q} - \omega_m (x_d i_{s,d} - \psi_p) \end{aligned} \tag{20.48}$$

where the permanent field flux $\psi_p$ represents the rotor circuit. The active and reactive power produced by the generator are as follows:

$$\begin{aligned} p_s &= v_{s,d} i_{s,d} + v_{s,q} i_{s,q} \\ q_s &= v_{s,q} i_{s,d} - v_{s,d} i_{s,q} \end{aligned} \tag{20.49}$$

The link between stator fluxes and generator currents:

$$\begin{aligned} \psi_{s,d} &= -x_d i_{s,d} + \psi_p \\ \psi_{s,q} &= -x_q i_{s,q} \end{aligned} \tag{20.50}$$

### Turbine and Machine Mechanical Equations

The generator motion equation is modeled as a single shaft, i.e., (20.27), as it is assumed that the converter controls are able to filter shaft dynamics. For the same reason, no tower shadow effect is considered in this model. In (20.27), the electrical torque is:

$$\tau_e = \psi_{s,d} i_{s,q} - \psi_{s,q} i_{s,d} \tag{20.51}$$

Substituting (20.50) in (20.51) leads to:

$$\tau_e = (\psi_{s,d} + (x_q - x_d) i_{s,d}) i_{s,q} \tag{20.52}$$

Mechanical equations are completed by the mechanical torque $\tau_t$ equation (20.25) and the turbine model (20.16) and (20.17)-(20.18) or (20.19)-(20.20) and the pitch angle control (20.21).

### VSC Regulators

The back-to-back VSC that connect the generator to the grid allow controlling four quantities. Controllable quantities are the converter currents $i_{s,d}$, $i_{s,q}$, $i_{c,d}$ and $i_{c,q}$. Typical controlled quantities are the active power injected into the grid, the grid side voltage, the dc voltage of the capacitor in the dc back-to-back connection, and the reactive power on the generator side. Several combinations have been proposed [3, 4, 119, 351]. A possible control scheme is as follows.

The currents on the generator side control the rotor speed and the generator reactive power:

$$\dot{i}_{s,q} = \frac{1}{T_{qs}}\left(\frac{p_w^*(\omega_m)}{\omega_m(\psi_p - x_d i_{s,d})} - i_{s,q}\right) \tag{20.53}$$
$$\dot{i}_{s,d} = (K_{ds}(q_{s0} - q_s) - i_{s,d})/T_{ds}$$

where $p_w^*(\omega_m)$ is the power-speed characteristic (20.22) or (20.23) and $q_{s0}$ is the reactive power determined at the initialization step.

The currents on the grid side control the active power and bus voltage:

$$\dot{i}_{c,q} = (K_{qc}(p_s - p_c) - i_{c,q})/T_{qc} \tag{20.54}$$
$$\dot{i}_{c,d} = (K_{dc}(v^{\mathrm{ref}} - v_h) - i_{c,d})/T_{dc}$$

The first of the previous equations indirectly models the dynamic of the dc system of the back-to-back connection. The steady-state error $p_s - p_c$ is a model of VSC and capacitor losses. A pure integrator can be used for modelling a loss-less back-to-back VSC:

$$\dot{i}_{c,q} = K_{qc}(p_s - p_c)/T_{qc} \tag{20.55}$$

If the dynamics of the VSC dc connection are fast, one can simply impose:

$$0 = p_s - p_c \tag{20.56}$$

## Limits

All currents in (20.53) and (20.54) undergo anti-windup limiters. The limits have to be carefully calculated to avoid overloading. On the dc side, one has:

$$\sqrt{i_{c,d}^2 + i_{c,q}^2} \le i^{\max} \tag{20.57}$$

where $i^{\max}$ is the VSC maximum current. Since (20.57) contains two variables, additional conditions are required to define $i_{c,d}^{\max}$ and $i_{c,q}^{\max}$. For example:

$$i_{c,q}^{\max} = i^{\max} \tag{20.58}$$
$$i_{c,d}^{\max} = \sqrt{(i^{\max})^2 - i_{c,q}^2}$$

Thus, one of the limits, e.g., $i_{c,d}^{\max}$ is a function of the other current. Then, for the lower limits:

$$i_{c,q}^{\min} = -i^{\max} \tag{20.59}$$
$$i_{c,d}^{\min} = -i_{c,d}^{\max}$$

Similar expressions can be defined for the limits of generator stator currents $i_{s,d}$ and $i_{s,q}$.

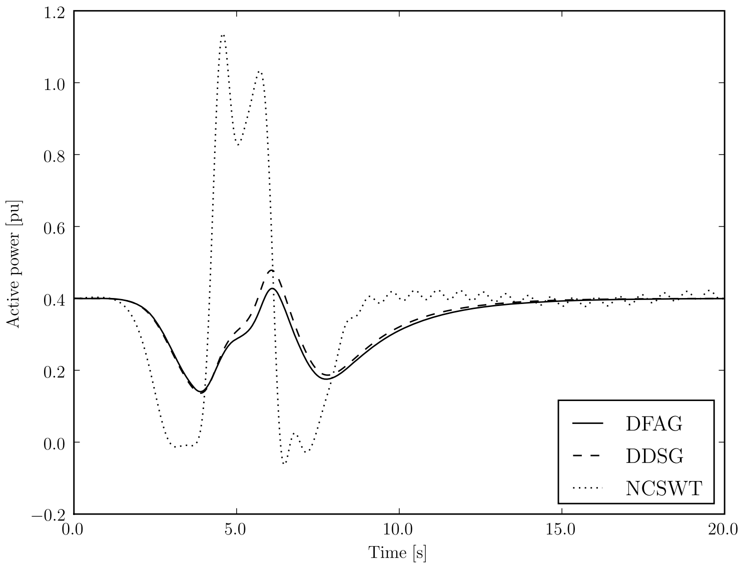


**Fig. 20.11** Comparison of transient behavior of different wind turbine types: doubly-fed asynchronous generator (DFAG), and direct-drive synchronous machine (DDSG), and non-controlled speed wind turbine (NCSWT)

### Example 20.4 Comparison of Wind Turbine Transient Behaviors

Figure 20.11 shows a comparison of the transient behavior of the three wind turbine types described in this section. In particular, the plot shows the power injected by the wind park at bus 2 of the IEEE 14-bus system.[1] The disturbance is a Mexican hat wavelet centered at $t = 5$ s, with a peak of 25 m/s. The initial power of the wind turbine is 0.4 pu and the wind park is composed of 40 machines (hence, the capacity of each machine is 1 MW). Mechanical data are the same for all wind turbine types. Machine and control data are provided in Appendix D.

The transient behaviors of the controlled speed wind turbines are quite similar: the controls of both the doubly-fed asynchronous generator and the direct-drive synchronous generator are able to slightly smooth the wind peak. On the other hand, the non-controlled wind turbine with squirrel-cage induction machine shows high oscillations that follows the wind peak and leads to undamped oscillations due to the shadow effect.

[1] The synchronous machine at bus 2 is substituted for a wind turbine.